Information Security

Unit -2 Cryptographic Algorithms

Unit- 2.1
Classical Cryptosystems

Compiled By Apil Adhikari MSc.CSIT
TU

Cryptography

- The word "Cryptography" came from two Greek word "Crypto" and "Graphy" meaning "Secret Writing" and it is the art or science of information hiding.

- Cryptography is a method of sharing or transmitting data in particular form so that only those for whom it is intended can read and process it.

- Mathematically, Cryptosystem can be defined as a 5-tuple

  (P, C, K, E, D)

  Where- P → Is a finite set of possible plain text

  C → Is a finite set of possible cipher text

  K → Is a key space, is a finite set of possible keys.

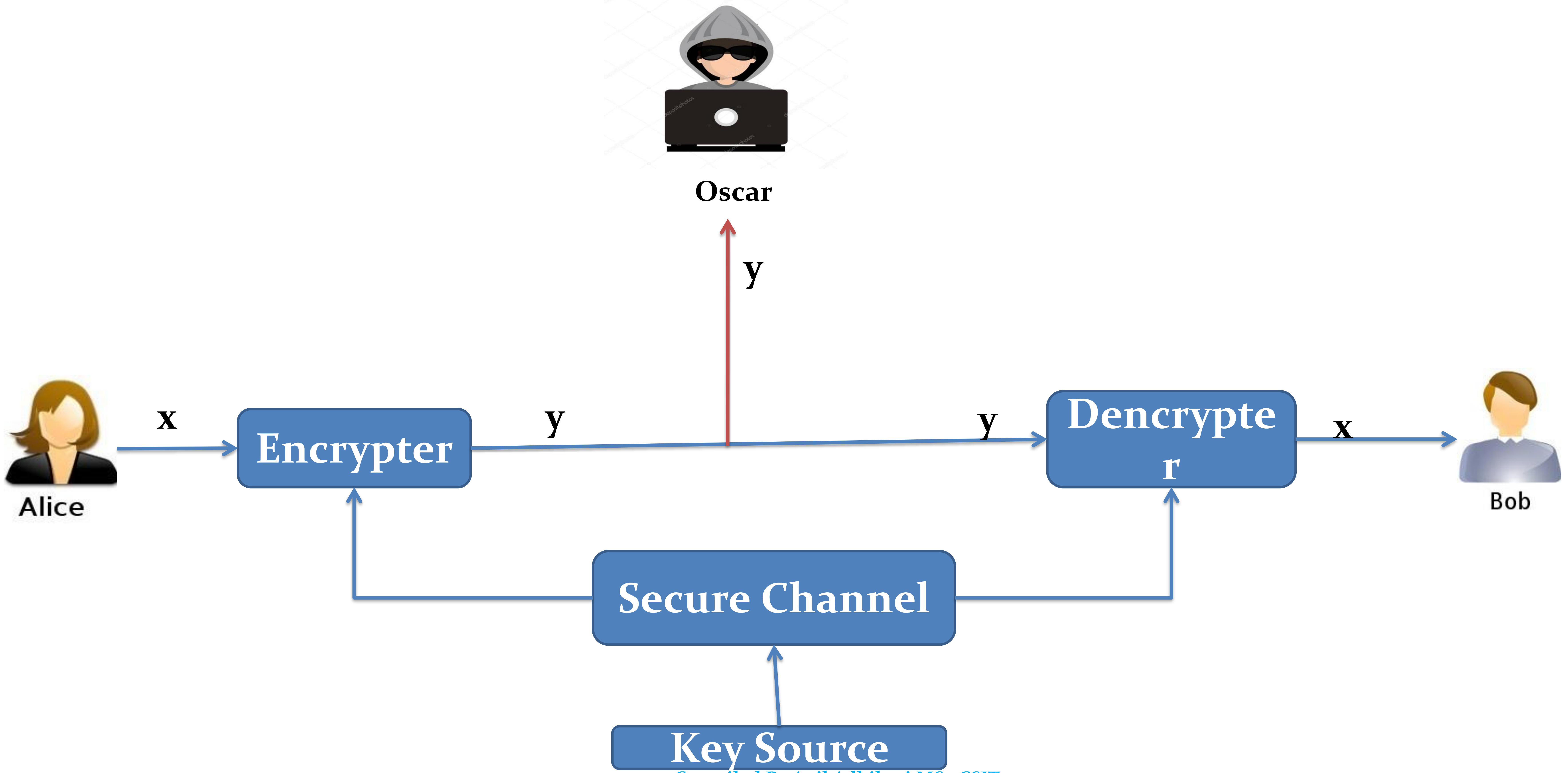
  For each $k \in K$ there is an encryption rule $e_k \in E$ and a corresponding decryption rule $d_k \in D$ such that,

  $e_k : P \rightarrow C$

  $d_k : C \rightarrow P$

  Such that $d_k(e_k(x)) = x$ for every plain text element $x \in P$.

Example of Simple Cryptosystem



- The fundamental objective of cryptography is to enable two people, usually referred to as Alice and Bob to communicate over an insecure channel in such a way that an opponent Oscar can not understand what is being said.
- This channel could be a telephone line or computer network.
- The information that Alice wants to send to Bob, which we call plaintext can be English text, numerical data or anything at all – its structure is completely arbitrary.
- Alice encrypts the plain text using a predetermined key and sends the resulting ciphertext over the channel.
- Oscar, upon seeing the ciphertext in the channel by eavesdropping, cannot determine what the plaintext was; but Bob, who knows the decryption key can decrypt the ciphertext and reconstruct the plaintext.

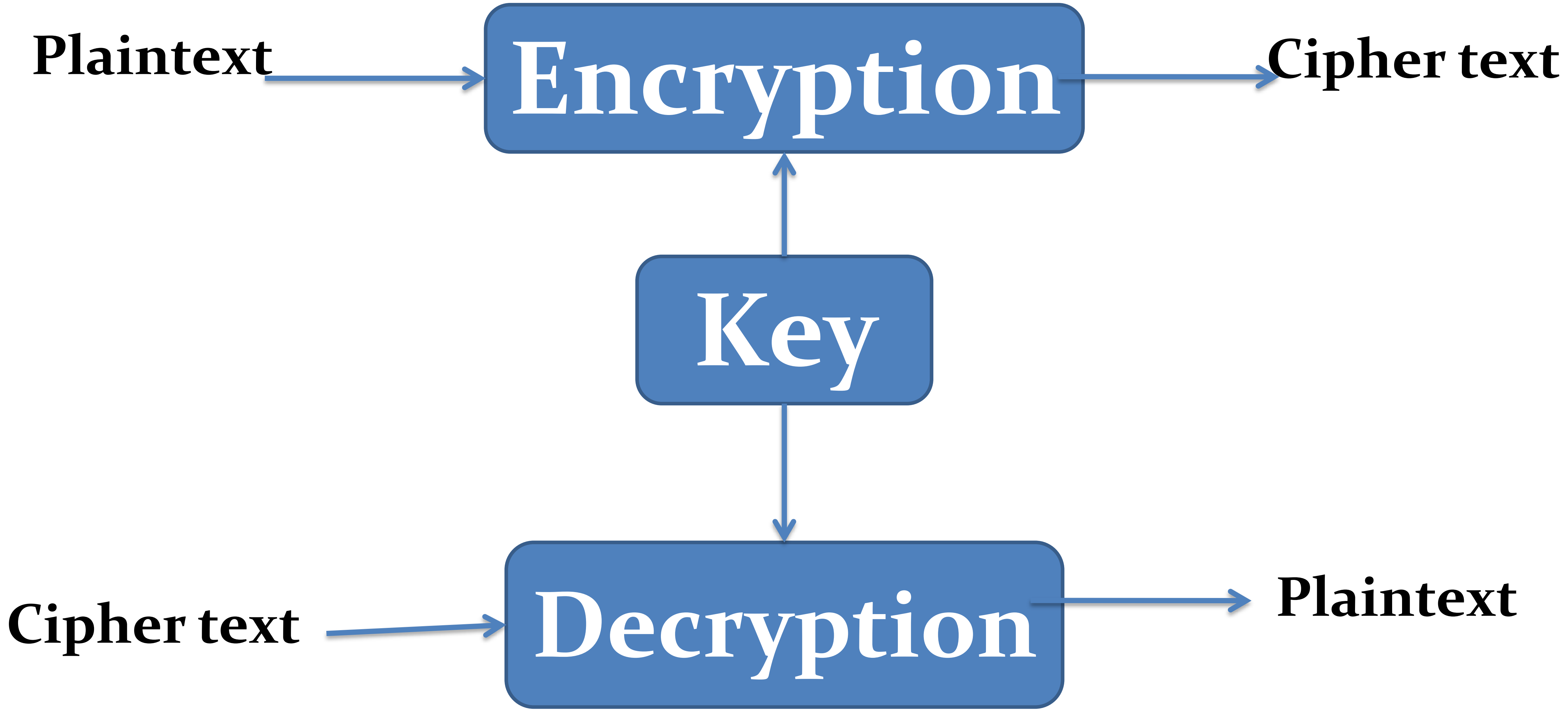
Key

- Key is a parameter or piece of information used to determine the output of cryptographic algorithm.
- A key is a binary number that is typically from 40-256 bit in length.

- While doing encryption, key determines the transformation of plaintext to the ciphertext and vice-versa.
- The data are encrypted or locked by combining the bits in the key mathematically with the data. At the receiving end, the key is used to unlock the code and restore the original data.

- In cryptosystem there are mainly two types of key are used and they are –
  1. Private key
  2. Public key

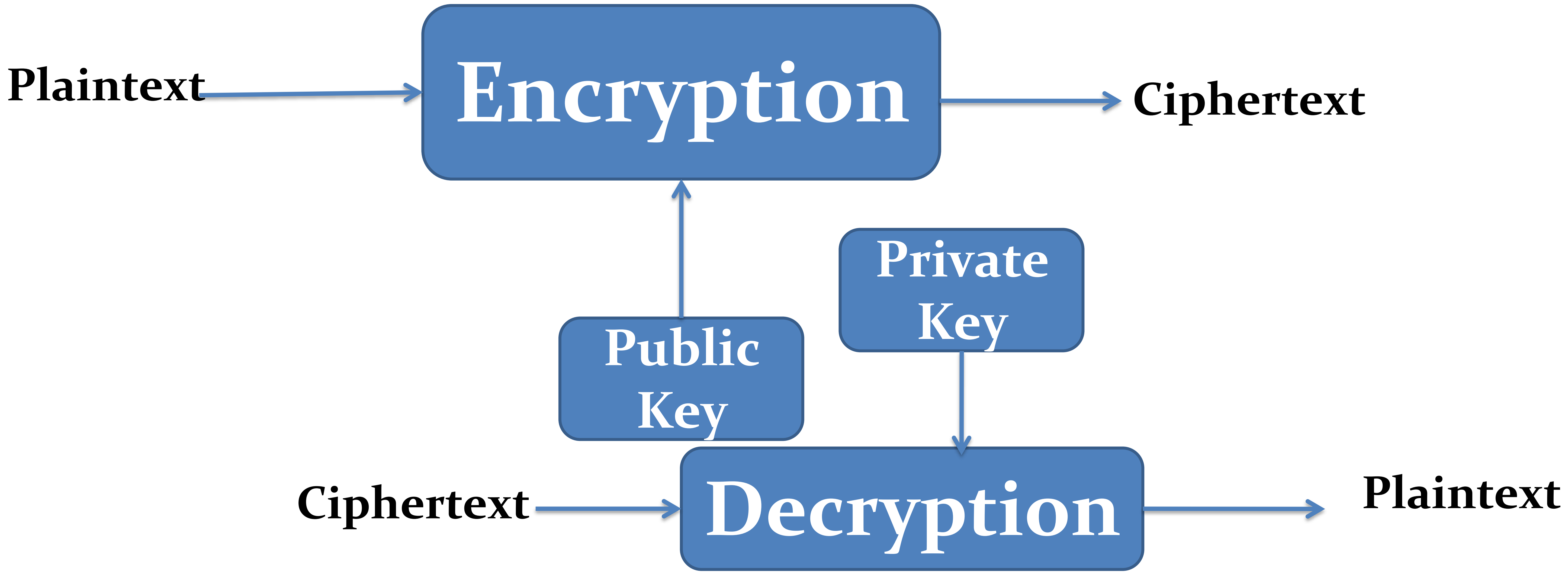
1. Private Key (Secret, Conventional, Symmetric)



- If both sender and receiver use the same key, the system is referred as symmetric or single or secret or conventional or private key encryption.

- For example, DES and AES algorithm

2. Public Key (Asymmetric, two key)



- If both sender and receiver use the different key, the system is referred as asymmetric or two key or public key encryption.

- For example, RSA and ELGamal algorithm

Classical Cryptosystems

1. Shift cipher :

- Shift cipher is a type of encryption where each letter in the plaintext is replaced by a letter a certain number of positions down the alphabet.
- The number of position to shift is determined by a key and it can be any integer from 1 to 25.

Shift cipher based on modular arithmetic:

Let $P = C = Z_{26}$ and for $0 <= k <= 25$, define

$e_k(x) = (x+k) \bmod 26$

and

$d_k(y) = (y-k) \bmod 26$

Where $(x,y) \in Z_{26}$

- For example, consider

  Plaintext: HELLOWORLD

  Key : 3

1. Encryption process ($e_k(x) = (x+k) \bmod 26$) :

   $e_k(H) = (7+3) \bmod 26 = 10 \Rightarrow K$

   $e_k(E) = (4+3) \bmod 26 = 7 \Rightarrow H$

   $e_k(L) = (11+3) \bmod 26 = 14 \Rightarrow O$

   $e_k(O) = (14+3) \bmod 26 = 17 \Rightarrow R$

   $e_k(W) = (22+3) \bmod 26 = 25 \Rightarrow Z$

   $e_k(R) = (17+3) \bmod 26 = 20 \Rightarrow U$

   $e_k(D) = (3+3) \bmod 26 = 6 \Rightarrow G$

   There for, Ciphertext = KHOORZRUOG

2. Decryption process ($d_k(y) = (y-k) \bmod 26$) :

   $d_k(K) = (10-3) \bmod 26 = 7 \Rightarrow H$

   $d_k(H) = (7-3) \bmod 26 = 4 \Rightarrow E$

   $d_k(O) = (14-3) \bmod 26 = 11 \Rightarrow L$

   $d_k(R) = (17-3) \bmod 26 = 14 \Rightarrow O$

   $d_k(Z) = (25-3) \bmod 26 = 22 \Rightarrow W$

   $d_k(U) = (20-3) \bmod 26 = 17 \Rightarrow R$

   $d_k(G) = (6-3) \bmod 26 = 3 \Rightarrow D$

   Thus, Plaintext = HELLOWORLD

2. Caesar Cipher:

- A Caesar cipher is a special case of shift cipher where each letter in plaintext is replaced by a key value K= 3.

3. Substitution Cipher :

- A substitution cipher is a type of encryption where each letter in the plaintext is replaced by a different letter or symbol according to some predetermined mapping. The mapping used is typically a permutation of the letters of the alphabets.

- Let $P = C = Z_{26}$ and K consists of all permutations of the 26 alphabet symbols 0,1,2......... 25.

- Now for each permutation $\pi \in K$ define

  $e_{\pi}(x) = \pi(x)$

  $d_{\pi}(y) = \pi^{-1}(y)$

  Where π^{-1} is the inverse permutation to π

3. Substitution Cipher :

- For example,
  - Let P = C A B
  - L = { A, B, C, D }
  - K = { ABCD, ACDB, ADCB, ADBC, ...}

- Let π = ADBC

This means:

A	B	C	D
A	D	B	C

Now the inverse permutation would be :

A	D	B	C
A	B	C	D

C	A	B	← Plaintext

$\downarrow \pi$

B	A	D	← Cipher text

$\downarrow \pi^{-1}$

C	A	B	← Plaintext

Monoalphabetic Cipher

- Monoalphabetic cipher is a type of encryption that uses a single fixed substitution method to encrypt a message.
- In this method each letter in a plaintext message is replaced by a corresponding letter in the ciphertext using a fixed mapping between the letters of the alphabets.
- For example, if the substitution key is A = D, B = E and so on then the plaintext letter "A" would always replaced by ciphertext letter "D" and so on.

- Here the main weakness of monoalphabetic cryptography is that it is vulnerable to frequency analysis attacks, where an attacker can analyze the frequency distribution of the letters in the ciphertext to determine the substitution key and recover the plaintext message.
- Examples-
  1. Shift cipher
  2. Caesar cipher
  3. Substitution cipher
  4. Affine cipher

Polyalphabetic Cipher

- Polyalphabetic cipher is a type of encryption that uses multiple substitution methods to encrypt a message.
- Unlike monoalphabetic cipher where a single substitution method is used for the entire message, polyalphabetic cipher uses different substitution method for different parts of the message.

  - Examples-
    1. Hill cipher
    2. Vigenere cipher
    3. Playfair cipher

4. Vigenere Cipher:

- Vigenere cipher is a type of polyaphabetic cipher that uses a keyword to encrypt and decrypt messages.
- To encrypt a message using the vigenere cipher, we start by choosing a keyword consisting of letters.
- For example, lets use the keyword "LEMON". Then we repeat the keyword over and over until it is the same length as the message we want to encrypt. For example, if we want to encrypt the message "ATTACKATDAWN" we repeat the keyword as -

"ATTACKATDAWN"	0	19	19	0	2	10	0	19	3	0	22	13
"LEMONLEMONLE"	11	4	12	14	13	11	4	12	14	13	11	4

- Next, we add the corresponding numerical values together using modulo arithmetic. i.e.

	0	19	19	0	2	10	0	19	3	0	22	13
+	11	4	12	14	13	11	4	12	14	13	11	4
	11	23	5	14	15	21	4	5	17	13	7	17
	L	X	F	O	P	V	E	F	R	N	H	R

Ciphertext

- To decrypt the message, we use the same keyword and repeat the above process but the only difference is we have to subtract the keyword from the ciphertext instead of adding.

5. Playfair Cipher:

- Invented by Charles Wheatstone in 1854 AD.
- It uses 5x5 matrix table for encryption and decryption.

1. Matrix Generation:

1. Choose keyword (eg. PLAYFAIRENCRYPTION)
2. Enter characters of keywords in 5x5 matrix row wise from left to right.
3. Fill remaining spaces in matrix with rest of English alphabets.
4. Combine i & j in the same cell.

P	L	A	Y	F
I/J	R	E	N	C
T	O	B	D	G
H	K	M	Q	S
U	V	W	X	Z

2. Encryption Process:

1. Break the plaintext in group of two alphabets
2. If both alphabets are same or only one is left add an "X" after first alphabet.
3. If both the alphabet in pair appears in the same row of matrix, replace them with alphabets to their immediate right respective.
4. If both the alphabet in pair appears in the same column, replace with alphabet immediately below them respectively.
5. If the alphabets are not in same row or column replace them with alphabets in the same row respectively, but at opposite pair of corners.

P	L	A	Y	F
I/J	R	E	N	C
T	O	B	D	G
H	K	M	Q	S
U	V	W	X	Z

Example-1: Plaintext = “LYEM”

Break the plaintext in a pair i.e.

L Y	E M
A F	B W

Ciphertext

Example-2: Plaintext = “NAMM”

Break the plaintext in a pair i.e.

N A	M X	M X
E Y	Q W	Q W

Ciphertext

P	L	A	Y	F
I/J	R	E	N	C
T	O	B	D	G
H	K	M	Q	S
U	V	W	X	Z

3. Decryption Process:

Example-1: C = "AFBW"

Break the plaintext in a pair i.e.

A F	B W	← Ciphertext
L Y	E M	← Plaintext

Example-2: C = "EYQWQW"

Break the plaintext in a pair i.e.

E Y	Q W	Q W	← Ciphertext
N A	M X	M X	← Plaintext

Transposition Cipher

- Transposition Cipher is a cryptographic algorithm where the order of alphabets in the plaintext is rearranged to form a cipher text.
- In contrast to substitution ciphers, which replace each plaintext character with a different character or symbol, transposition ciphers simply shuffle the positions of the characters within the message.

  - Examples-
    1. Columnar cipher
    2. Rail Fence Cipher

6. Rail Fence Cipher:

- The Rail Fence Cipher is a simple transposition cipher that rearranges the characters of a plaintext by writing them in a zigzag pattern across a specified number of "rails" or lines.
- This technique creates an encrypted message by reading off the characters row by row from the zigzag pattern.

1. Encryption:

1. Write the plaintext message horizontally across a number of "rails" or lines.
2. As you write the message, alternate the direction of the successive character, moving diagonally upwards and downwards.
3. The number of rails used determines the depth of the zigzag pattern.
4. Once the entire message is written out, read off the characters row by row to form the ciphertext.

2. Decryption:

1. To decrypt the ciphertext, repeat the process of writing out the zigzag pattern with the same number of rails used for encryption.

2. As you write the zigzag pattern, mark the positions where the characters from the ciphertext would be placed.

3. Once all positions are marked, read off the characters row by row to reconstruct the original plaintext message.

Example:

Plaintext : "ATTACKATONCE"

Depth : 2

1. Encryption:

A		T		C		A		O		C	
	T		A		K		T		N		E

Ciphertext: A T C A O C T A K T N E

2. Decryption :

* A		* T		* C		* A		* O		* C	
	* T		* A		* K		* T		* N		* E

Plaintext: A T T A C K A T O N C E

Assignment

Q1) Show encryption and decryption of " BCACOURSE" using Caesar cipher.

Q2) Show encryption and decryption of "HELL" using the key "AI" using Vignere cipher.

Q3) The message "IMOGUN" was encrypted with a Playfair cipher using keyword "GALOIS". Decrypt the message.

Q4) Given a message M = "BCAPROGRAMISAHOTCAKE", encrypt M using Rail Fence cipher with rail size 3.

Q5) Using Rail fence cipher encrypt the text "LEARNING AND TEACHING ARE DIFFERENT" using 3 as rails.

!!!Thank you!!!